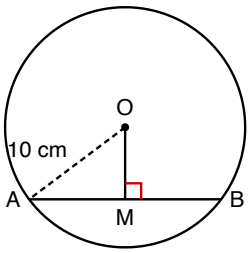
Shape and Space 6 — Chords · Alternate segment theorem · Intersecting chords

Calculator allowed. Marks in brackets. ★ = harder question. Diagrams are not drawn accurately. Give a reason for every angle fact you use.

1 O is the centre of a circle of radius 10 cm. AB is a chord of length 16 cm and M is the mid-point of AB. [3]

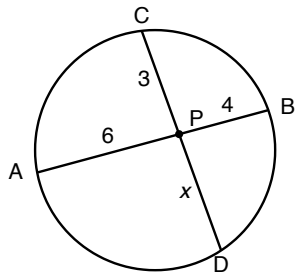


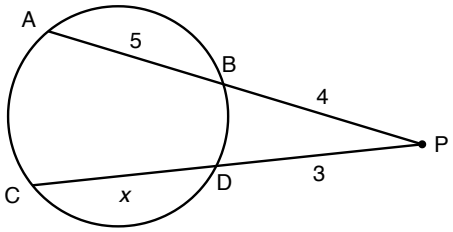
a Write down the length of OA. **b** Write down the length of AM.

c Work out the length of OM.

a OA is a radius: 10 cm. **b** The line from the centre to the mid-point of a chord is perpendicular to it and bisects it: AM $= 8$ cm. **c** Pythagoras in the right-angled triangle OMA: $\mathrm{OM} = \sqrt{10^2 - 8^2} = \sqrt{36} = 6$ cm.

2 In each diagram find x. The lengths are in cm. [6]



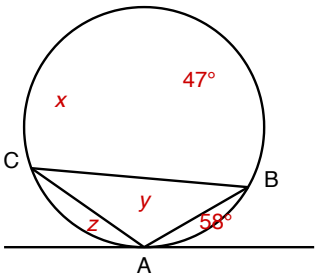


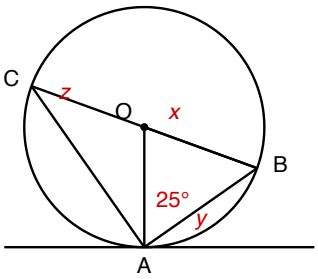
a The chords AB and CD cross at P inside the circle. AP $= 6$, PB $= 4$, CP $= 3$, PD $= x$.

b P is outside the circle. PB $= 4$, BA $= 5$, PD $= 3$, DC $= x$.

> **a** Intersecting chords: $\mathbf{AP} \times \mathbf{PB} = \mathbf{CP} \times \mathbf{PD}$, so $6 \times 4 = 3x \Rightarrow x = 8$ cm.
>
> **b** The same rule with P outside — measure every length *from P*: $\mathbf{PB} \times \mathbf{PA} = \mathbf{PD} \times \mathbf{PC}$. $\mathbf{PA} = 4 + 5 = 9$ and $\mathbf{PC} = 3 + x$: $4 \times 9 = 3(3 + x) \Rightarrow 36 = 9 + 3x \Rightarrow x = 9$ cm.

3 In each diagram find the angles x, y and z. Give a reason for each step. [12]





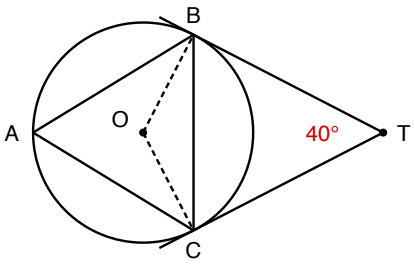
a The line through A is a tangent. The angle between the tangent and the chord AB is 58°, and angle ABC $= 47^\circ$.

b O is the centre and the line through A is a tangent. Angle OAB $= 25^\circ$.

a $x = \angle ACB = 58^\circ$ – the angle between a tangent and a chord equals the angle in the alternate segment. $y = \angle BAC = 180^\circ - 58^\circ - 47^\circ = 75^\circ$ – angles in a triangle add up to 180°. $z = \angle ABC = 47^\circ$ – alternate segment theorem again, now for the chord AC (the angle between the tangent and AC equals the angle in the alternate segment).

b Triangle OAB is isosceles (OA = OB, radii), so $\angle OBA = 25^\circ$ and $x = \angle AOB = 180^\circ - 50^\circ = 130^\circ$. $y = 90^\circ - 25^\circ = 65^\circ$ – a tangent is perpendicular to the radius at the point of contact. $z = \angle ACB = 65^\circ$ – the angle at the centre is twice the angle at the circumference ($130^\circ \div 2$), or equally by the alternate segment theorem, $z = y$.

4 TB and TC are tangents from T to the circle with centre O. A is a point on the major arc BC. Angle BTC $= 40^\circ$. [4]



a Find angle TBC. **b** Find angle BAC. **c** Find angle BOC.

a Two tangents from the same point are equal, TB = TC, so triangle TBC is isosceles: $\angle TBC = \frac{1}{2}(180^\circ - 40^\circ) = 70^\circ$.

b TB is a tangent and BC a chord, so by the alternate segment theorem $\angle BAC = \angle TBC = 70^\circ$.

c Angle at the centre is twice the angle at the circumference: $\angle BOC = 2 \times 70^\circ = 140^\circ$. (Check: in quadrilateral OBTC the angles at B and C are 90° each, so $\angle BOC = 360^\circ - 90^\circ - 90^\circ - 40^\circ = 140^\circ$ ✓.)

Total: 25 marks

Check: perpendicular from the centre bisects the chord (then Pythagoras) · alternate segment: tangent–chord angle = angle in the opposite segment · intersecting chords, inside or outside: $\mathrm{AP} \times \mathrm{PB} = \mathrm{CP} \times \mathrm{PD}$ with every length measured from the crossing point P · tangent $\perp$ radius · two tangents from a point are equal.